SECTION 5 · SUBSECTION 5.1
FILE CATALOGUE: BACKEND

5.1.1 INTRODUCTION.

This catalogue covers every tracked file at the analysed commit, plus two local files that are present but not yet tracked. It excludes generated and vendored material: the virtual environment, *node_modules*, build output, Chroma persistence, run outputs, caches and this handbook. Line counts are non-empty lines at the working tree, not logical statements. The purpose is orientation: a reader should be able to name the file that produces any value seen in the output (figure 5.1-1).

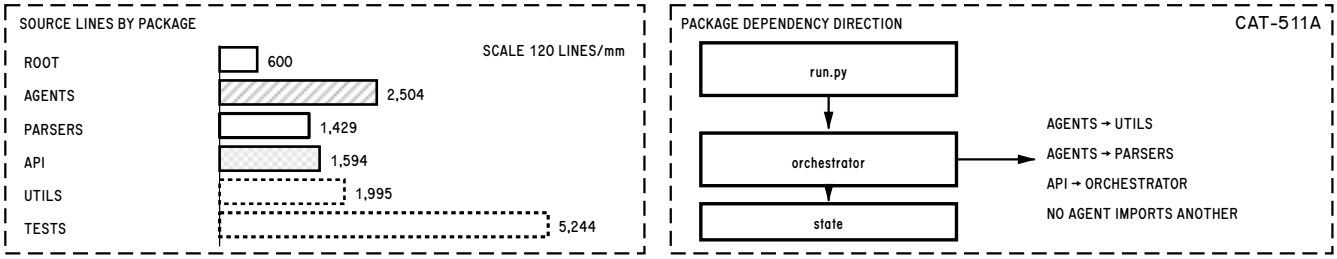


Figure 5.1-1. Package Size and Allowed Import Direction

5.1.2 BACKEND MODULES.

Thirty-three Python modules carry the backend. *src/parsers* turns syntax into dataclasses; *src/utils* supplies everything that talks to the outside world; *src/agents* holds the seven stages; *src/api* adds transport and persistence. Empty *__init__.py* files are omitted from the table.

Table 5.1-1. Backend Module Inventory (empty lines excluded)

Module	Lines	Responsibility
run.py	204	Argument parsing, size pre-check, pipeline call, ten save calls
src/orchestrator.py	335	Phase order, three-thread join, coupling rebind, output writers
src/state.py	61	*FileMetadata* and *ArchaeonState*; the only cross-agent contract
agents/ingestion.py	135	Filter, sort, tier, cap at 3,000, batch fetch, line counts
agents/ast_parser.py	335	Parallel parse, language dispatch, import internals, parse errors
agents/dependency_graph.py	254	Nodes, edges, cycles, PageRank, topological order, per-file stats
agents/complexity_scorer.py	353	Function scores, thresholds, risk reasons, deferred coupling
agents/code_rag.py	254	Chunking, batched embeddings, Chroma write, raw-text release
agents/explainability.py	373	File ranking, context budget, prompt, chat call, explanation cache
agents/doc_generator.py	800	Onboarding, agent context, explanations, complexity JSON; section guards
api/main.py	322	Dual route families, CORS, headers, request validation, queue choice
api/models.py	142	Pydantic request, status, summary and result contracts
api/pipeline_runner.py	295	API phase sequence, progress marks, result serialisation, failure record
api/job_store.py	552	Memory and Redis stores, WATCH loop, active-index reconciliation
api/job_store_migrate.py	95	Operator command to rebuild and arm the active index (untracked)
api/queue.py	107	Redis connection, RQ queue, enqueue, depth and worker statistics
api/worker.py	81	Health listener, Redis probe, RQ foreground worker
parsers/base.py	91	*FunctionInfo*, *ClassInfo*, *ImportInfo*, *SymbolTable*, *ComplexityScore*
parsers/python_parser.py	354	Python functions, classes, decorators, imports, docstrings
parsers/js_parser.py	285	JavaScript and TypeScript declarations, arrows, imports
parsers/generic_parser.py	426	Go, Rust, Java, C and C++ symbol extraction
parsers/complexity.py	273	Python and JavaScript branch counting; generic fallback
utils/github_api.py	197	URL parsing, metadata, tree, raw fetch pool, API fallback
utils/filters.py	109	Extension allow-list, 24 excluded directories, 150 KB ceiling
utils/tree_sitter_utils.py	88	Language grammar loading and parser cache
utils/chunker.py	223	Module, class and function chunks with stable identifiers
utils/embedder.py	114	Embedding client, batching, passage and query input types
utils/retriever.py	233	Collection naming, cosine query, metadata filters, parsing
utils/nvidia_client.py	208	Chat client, timeout, two attempts, output sanitisation
utils/explanation_cache.py	135	Content-addressed explanation cache on local disk
utils/diff_engine.py	242	SHA-based manifest diff and state rehydration
utils/graph_utils.py	387	Per-language import resolution and pyvis HTML export
utils/metrics.py	59	Phase timing and resident-memory prints

NOTE

Two files are outside the commit. *job_store_migrate.py* and *test_active_index.py* exist only in the working tree and are marked accordingly. Their presence means the committed source cannot rebuild the active index, even though the store can fall back to scanning records.

CATALOGUE CHECK	REVIEWER
√ COUNT - 33 modules named	
√ LINES - non-empty, working tree	
1 EXCLUDED - env, deps, output, caches	
2 MARKED - untracked files	

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